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DESIGN OF A LEAN PREMIXED PREVAPORIZED CAN COMBUSTOR

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ABSTRACT

Increasingly, more stringent emissions regulations have necessitated new gas turbine combustor designs with low pollutant emissions. Radically different modern designs have been developed to meet these requirements while maintaining high combustion efficiencies and good flame stability. While several published methodologies for conventional combustor design exist, none exist for modern ones.

This paper describes the development of a new preliminary design algorithm for a modern lean premixed prevaporized (LPP) combustor. It also introduces a new LPP combustor concept. The approach used is multi-disciplinary in nature, applying empirical and semi-empirical models in the algorithm to capture complex processes such as droplet evaporation, chemical reaction, jet mixing, and heat transfer. The resulting set of procedures allows a designer to quickly define the detailed geometry of the combustor and provides an assessment of its performance.

The preliminary design procedures were verified using the advanced numerical techniques of computational fluid dynamics (CFD). Reasonable agreement between predictions from the preliminary design and numerical analysis was achieved which indicated that the design procedures have been developed successfully.

NOMENCLATURE

$\left(\frac{A}{F}\right)$	Air-to-fuel ratio by mass
A	Area
AR	Area ratio or aspect ratio
c	Chord
C_D	Discharge coefficient
C_p	Static pressure recovery coefficient
C_s	Flameholder shape factor
D	Diameter
D_c	Characteristic dimension of flameholder
e	Trip strip height.
J	Momentum-flux ratio
k	Proportionality constant
K	Jet-to-annulus dynamic pressure ratio upstream of holes
K_{sw}	Swirler blade factor
L	Length
$\dot{m}$	Mass flow rate
n	Quantity
p	Trip strip spacing.
P	Pressure
q	Dynamic pressure
Q	Volumetric flow rate or pattern factor
R	Ideal gas constant
S_L	Laminar flame speed
S_n	Swirl number
t	Thickness or time
T	Temperature
V	Velocity
W	Air mass fraction
Y_{max}	Maximum jet penetration distance

Greek Symbols

α_0	Thermal diffusivity upstream of flameholder
ΔP	Pressure loss
ΔP_{3-4}	Overall pressure loss
ΔT	Temperature rise.
ϕ	Equivalence ratio, $\left(\frac{A}{F}\right)_{stoic} / \left(\frac{A}{F}\right)_{act}$
ρ	Density
σ	Solidity
θ	Wall or blade angle

Subscripts

3	Combustor inlet
4	Combustor outlet

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a	Property of air
an	Annulus value
BO	Blowout value
c	Value for a conventional combustor
ctr	Contraction value
d	Value downstream of swirler
$diff$	Diffuser value
do	Dump diffuser value
dp	Dump diffuser value
DZ	Dilution zone value
f	Fuel value
fl	Flame value
g	Hot combustion gas value
h	Dilution hole value
hub	Hub value
$hub,1$	Mixer hub value
$hub,2$	Combustor swirler hub value
ig	Ignition value
j	Dilution jets value
l	Liquid fuel value
L	Liner value
$L,2$	Liner value downstream of contraction
max	Maximum value
mix,i	Inner mixer swirler value
mix,o	Outer mixer swirler value
n	Nozzle value
$overall$	Overall value
p	Value for a premixed combustor
pd	Prediffuser value
pl	Plenum value
pm	Premixer value
PZ	Primary zone value
ref	Reference/casing value
$ref,1$	Casing value upstream of dome
$ref,2$	Casing value downstream of contraction
st	Stoichiometric value
sw	Swirler value
$swirl$	Combustor swirler value
u	Unburned gas value
v	Swirler vane

INTRODUCTION

Governmental regulations for gas turbine emissions are becoming increasingly more stringent. The demand for new improved designs with ultra-low pollutant emissions is rapidly moving to the forefront of combustor development. Radical modern combustor designs have emerged to achieve emission requirements while maintaining the high combustion efficiency and good flame stability characteristics of conventional combustors. Lean premixed (LP) for gaseous and lean premixed prevaporized (LPP) for liquid fuelled engines are two modern designs proven successful at meeting governmental regulations.

Design of these new modern LP and LPP combustors for gas turbines is complicated by several factors. First, detailed information about current designs and the tools developed for their design are regarded as proprietary by large engine manufacturers. Additionally, there exists no published detailed methodology on their design since they are relatively new technology and each design tends to be drastically different.

Several authors have published preliminary designs for conventional aircraft gas turbine combustors [1–3]. However, they do not apply well to modern premixed ones. Lefebvre [4] and Mellor [5] have compiled extensive lists of empirical, semi-empirical, and analytical tools. While specifically addressing the design of conventional aircraft combustors, the authors also provide information and references for new modern combustors.

Using a multidisciplinary approach, a methodology for the preliminary design of a LPP combustor was formulated. These procedures are the initial step in the overall design process which precede complex numerical analysis using computational fluid dynamics (CFD) and finite element analysis (FEA). The developed methodology provides a detailed set of geometry and an estimate of its performance.

This paper describes the basic elements of the preliminary design procedure and summarizes the results of its verification. A detailed description including its application and verification is presented elsewhere [6].

APPROACH

Figure 1 illustrates the LPP concept, which uses a can configuration. In this particular implementation, air discharged from the compressor enters the combustor at Station 3.0. The air passes through a faired prediffuser before it is dumped into the annulus. In the region just downstream of the sudden expansion, the cross-sectional flow area in the annulus is large to ensure that any asymmetry in the flow is avoided.

As the air accelerates through the annulus, a portion is admitted by a row of holes in the liner to dilute the hot combustion gases. The other portion of the air flows through the annulus where it cools the outside of the liner wall. One characteristic of modern premixed combustion is that less air is available for liner wall cooling. Therefore, wall cooling is enhanced using two different devices: trip strips on the cold side of the liner wall and a thermal barrier coating (TBC) on the hot side.

The annulus air used for combustion is dumped into a plenum that supplies the premixer. Inside the premixer, this air passes through two concentric, counter-swirling axial swirlers. Liquid fuel is injected into the resultant highly sheared flow as a fine droplet spray where it evaporates and forms fuel vapour. This vapour mixes with the surrounding air to produce a combustible mixture.

A third axial swirler is used to induce a strong swirl component to the fresh fuel/air mixture as it enters the liner for combustion. This ensures the formation of a strong recirculation vortex inside the liner and promotes stable combustion.

Operating Conditions. The combustor must operate stably and efficiently with low pollutant emissions over a wide range of operating conditions. These operating conditions are known prior to performing the combustor design through thermodynamic engine calculations or experimental testing.

For simplicity, the static temperature and pressure everywhere inside the combustor are assumed equal to the stagnation values. This is a reasonable assumption since the Mach number everywhere is below 0.3. Thus, the following conditions are

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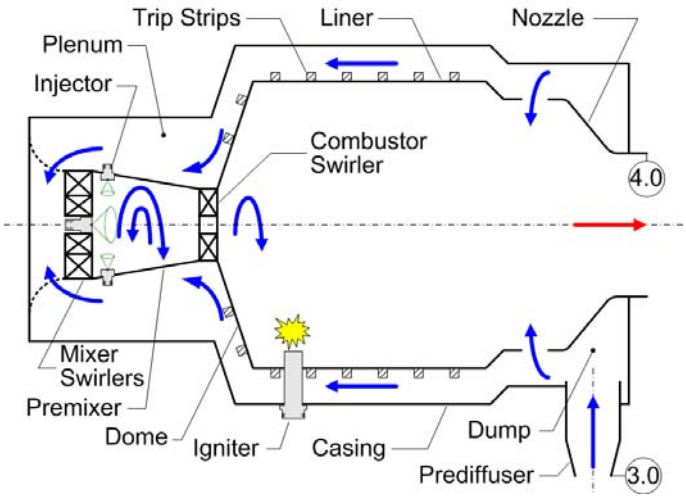

Figure 1. LPP CONCEPT.

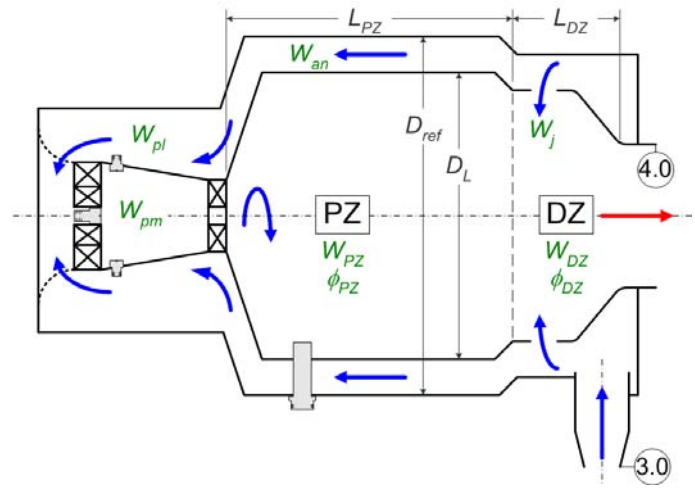

Figure 2. AIRFLOW DISTRIBUTION AND MAJOR DIMENSIONS.

known for each operating point: the air flow rate $\dot{m}_3$, the inlet temperature T_3, the outlet temperature T_4, and the inlet pressure P_3. Additionally, the density of the air at the inlet is given by the ideal gas law:

$$\rho_3 = \frac{P_3}{R_a T_3} \tag{1}$$

Equivalence Ratio. The overall equivalence ratio, $\phi_{overall}$, necessary to achieve the specified temperature rise $(\Delta T = T_4 - T_3)$ is estimated by assuming equilibrium combustion at constant pressure. While the combustion process will not generally achieve equilibrium in practice, the assumption is valid since the heat release, and consequentially the formation of CO, occurs fast when compared to other reactions such as the formation of NO_x.

The equilibrium constant approach, discussed in detail by Turns [7], is used to compute $\phi_{overall}$. The overall equivalence ratio is related to the operating conditions and fuel flow as follows:

$$\phi_{overall} = \frac{\left(\frac{A}{F}\right)_{st}}{\left(\frac{\dot{m}_3}{\dot{m}_f}\right)} \tag{2}$$

Airflow Distribution. The airflow distribution is selected to provide the desired stoichiometry in the primary and dilution zones, which are illustrated in Fig. 2. As a result, the fractions of the total air flow in the various combustor passages, W_i, are resolved.

Premixed combustors operating below 1900 K were determined to exhibit little NO_x formation if the fuel and air was uniformly mixed [8]. A lower flame temperature of 1800 K is selected to minimize the peak value in any local hot spots caused by unmixedness, which is present in all LPP combustors. Inherent issues with combustor stability and excessive CO are expected if operating any closer to the weak extinction limit.

The primary zone equivalence ratio, ϕ_{PZ}, is computed using the procedures discussed in the previous section. It is set to provide the desired flame temperature of $T_{fl} = 1800$ K. Furthermore, the dilution zone equivalence ratio is equal to the overall equivalence ratio. Hence:

$$\phi_{DZ} = \phi_{overall} \tag{3}$$

$$\phi_{PZ} = f\left(T_{fl}\right) \tag{4}$$

The necessary air fractions in the main combustor zones (i.e., primary and secondary zone) and in the combustor passages are:

$$W_{PZ} = \frac{\phi_{overall}}{\phi_{PZ}} \tag{5}$$

$$W_{DZ} = \frac{\phi_{overall}}{\phi_{DZ}} \tag{6}$$

$$W_{pl} = W_{pm} = W_{PZ} \tag{7}$$

$$W_j = W_{DZ} - W_{PZ} \tag{8}$$

$$W_{an} = 1.0 - W_j \tag{9}$$

Combustor Sizing. Combustor sizing refers to the definition of the reference (casing) diameter, D_{ref}, to provide sufficient stability without incurring excessive pressure losses. The reference diameter, illustrated in Fig. 2, defines the total ideal flow area inside the combustor. All operating points (i.e., idle, full power) are considered and the smallest size that provides stability over the entire range of operation is chosen.

Specification of the size for a premixed combustor is difficult as design data is not currently available. Thus, the premixed combustor is scaled from an equivalent conventional combustor. The scaling parameter chosen for this algorithm is the lean blowout velocity, which is related to flame stability.

The casing size for most industrial combustors is dictated by the overall pressure loss, ΔP_{3-4}. For conventional combustors, the cross-sectional area of the casing is determined from [9]:

$$A_{ref} = \left[\frac{R_a}{2}\left(\frac{\dot{m}_3 T_3^{0.5}}{P_3}\right)^2 \left(\frac{\Delta P_{3-4}}{q_{ref}}\right)\left(\frac{\Delta P_{3-4}}{P_3}\right)^{-1}\right]^{0.5} \tag{10}$$

Suitable values for $\Delta P_{3-4}/P_3$ and $\Delta P_{3-4}/q_{ref}$ are chosen by the designer based on data provided by Lefebvre [9].

The liner diameter must be chosen carefully. It would appear desirable to select the largest diameter possible and reduce the velocity of the flow within the liner. This increases the residence

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time in the combustor and promotes stability. However, increasing the liner diameter for a fixed casing diameter results in a reduction of the annulus flow area A_{an}. The smaller area results in higher annulus velocities that decreases the static pressure in the annulus. Therefore, a larger liner diameter is undesirable since a high static pressure drop across the liner admission holes is necessary to provide adequate penetration of the jets. Sawyer [1] suggests that A_L/A_{ref} should be kept within 0.60-0.72 for conventional combustors.

The combustor is resized for premixed combustion to maintain the stability of an equivalent conventional combustor [10]. The scaling parameter used here to measure stability is the lean blowout velocity. For a bluff-body flame holder, the lean blowout velocity for homogeneous fuel-air mixtures is defined as follows [11]:

$$V_{B0} = C_s \left(D_c {S_L}^2 / \alpha_0 \right) \tag{11}$$

The diameter of the combustion chamber, D_L, is chosen as the characteristic dimension, D_c.

Non-dimensionalizing Eqn. (11) with the average axial velocity through the liner gives:

$$\frac{V_{B0}}{V_L} = \frac{\pi}{4} \frac{C_s \rho_g D_L^3 S_L^2}{\alpha_0 \dot{m}_L} \tag{12}$$

where ρ_g is the density of the hot combustion gases inside the liner.

The shape factor C_s is a value specific to the flameholding device, which is the swirler in this case. It can be assumed to remain constant for both the conventional and premixed cases. Eqn. (12) becomes:

$$\frac{V_{B0}}{V_L} \propto \frac{\rho_g D_L^3 S_L^2}{\alpha_0 \dot{m}_L} \tag{13}$$

where the density obeys the ideal gas law.

It is reasonable to assume that the flame temperature varies linearly with the equivalence ratio for very weak mixtures. Noting that α_0 and $\dot{m}_L$ remain constant, Eqn. (13) becomes:

$$\frac{V_{B0}}{V_L} \propto \frac{D_L^3 S_L^2}{\phi_{PZ}} \tag{14}$$

To maintain the stability of the original conventional combustor, the following must be satisfied:

$$\left(\frac{V_{B0}}{V_L} \right)_c = \left(\frac{V_{B0}}{V_L} \right)_p \tag{15}$$

where c designates the value for the conventional combustor and p for the premixed combustor.

Substituting Eqn. (14) into Eqn. (15) and rearranging yields the diameter of the liner necessary for premixed combustion:

$$\frac{D_{L_p}}{D_{L_c}} = \left(\frac{\phi_{PZ_p}}{\phi_{PZ_c}} \right)^{\frac{1}{3}} \left(\frac{S_{L_c}}{S_{L_p}} \right)^{\frac{2}{3}} \tag{16}$$

The laminar flame speed, the maximum homogeneous burning velocity, is determined using the modified Semenov equation [12]. The application of Eqn. (16) yields a scaling factor which is consistent with existing premixed combustors. Leonard and Stegmaier [8] stated that the required combustion volume for stable premixed operation is roughly twice that of a conventional combustor.

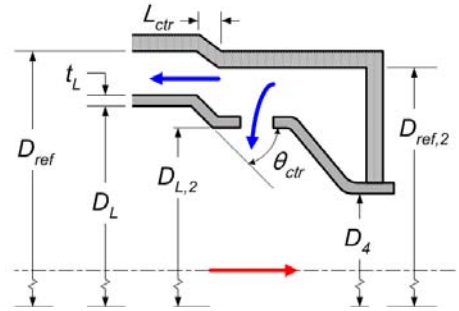

Figure 3. CONTRACTION DIMENSIONS.

The casing diameter of the premixed combustor is selected to provide the desired flow velocities in the annulus. In addition to the annulus flow considerations discussed previously, the high velocities desirable for augmented backside convection (ABC) cooling of the liner must also be considered.

Contraction. A small contraction in the aft section of the liner is used to accelerate the flow and end the recirculation zone (Fig. 3). A 30 percent increase in velocity is chosen based on a survey of many existing designs. The casing must contract as well to maintain a constant annulus area.

The liner diameter, casing diameter, and the length of the contracted section:

$$D_{L,2} = \sqrt{0.7} D_L \tag{17}$$

$$D_{ref,2} = \sqrt{\left(D_{L,2} + 2t_L\right)^2 + \frac{4}{\pi} A_{an}} \tag{18}$$

$$L_{ctr} = \frac{D_L - D_{L,2}}{2 \tan \theta_{ctr}} \tag{19}$$

Diffuser Design. The diffuser, illustrated in Fig. 4, consists of a conical prediffuser followed by a sudden expansion. The optimum geometry for these components is selected to minimize the overall pressure loss incurred across the diffuser. This overall loss is equal to the sum of the losses for each component:

$$\Delta P_{diff} = \Delta P_{pd} + \Delta P_{dp} \tag{20}$$

The prediffuser is designed to provide a 60 percent reduction in velocity based on the recommendations of Lefebvre [9]. From this requirement, the prediffuser geometry is defined by the following relations:

$$A_{pd} = AR_{pd} \cdot A_3 \tag{21}$$

$$D_{pd} = \sqrt{\frac{4}{\pi} A_{pd}} \tag{22}$$

$$L_{pd} = \frac{D_{pd} - D_3}{2 \tan \theta_{pd}} \tag{23}$$

where AR_{pd} is the prescribed prediffuser outlet-to-inlet area ratio.

The wall angle, θ_{pd}, is chosen to minimize the pressure loss incurred across the prediffuser. This loss is given by [9]:

$$\Delta P_{pd} = \left[\left(1 - \frac{1}{AR_{pd}^2} \right) - C_p \right] \cdot q_3 \tag{24}$$

where q_3 is the inlet dynamic pressure and C_p is the static pres-

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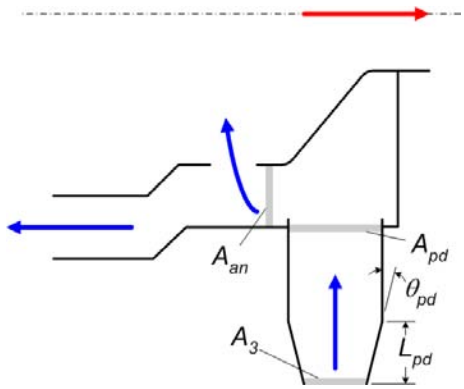

Figure 4. DIFFUSER GEOMETRY.

sure recovery coefficient.

The static pressure recovery coefficient is a measure of the diffuser effectiveness at converting kinetic energy into static pressure. It is estimated based on the diffuser dimensions using the widely accepted performance chart for conical diffusers created by Sovran and Klomp [13].

The pressure losses across the sudden expansion correspond well with the theoretical dumping losses [2]. Applying the momentum equation assuming uniform, steady, incompressible, and frictionless flow:

$$\Delta P_{dp} = \left[1 - \frac{1}{AR_{dp}}\right]^2 \cdot q_{pd} \tag{25}$$

where AR_{dp} is the outlet-to-inlet area ratio of the sudden expansion and q_{pd} is the prediffuser exit dynamic pressure.

Computation of the dump area ratio AR_{dp} has proven to be somewhat problematic. The flow in the dump region is highly separated and an effective flow area is difficult to define. It is therefore assumed in the present design that the flow reattaches in the annulus. Thus, the dump diffuser area ratio is given by:

$$AR_{dp} = \frac{A_{an}}{A_{pd}} \tag{26}$$

Premixer Design. The premixer is designed to evaporate the liquid fuel and mix the fuel vapour with air. Figure 5 illustrates the premixer concept and indicates the relevant parameters.

Precautions must be taken to ensure that the fuel/air mixture does not react prior to entering the primary zone of the combustor. Any potential flameholders such as struts, fuel lines, hubs, or any other bluff bodies where combustion can occur in its wake must be eliminated. Therefore, the co-axial swirlers are designed to ensure that no recirculation occurs inside the premixer [14].

Slow moving boundary layers on the premix tube walls must also be minimized to ensure that the flame will not creep upstream and cause flashback. The premixer tube converges to continually accelerate the flow and flatten any thick, developed boundary layers.

Chemical reaction can take place inside the premixer by autoignition after exposure to heated air. This spontaneous ignition can be avoided by ensuring that the residence time of the fuel/air mixture in the premixer is less than the autoignition delay time.

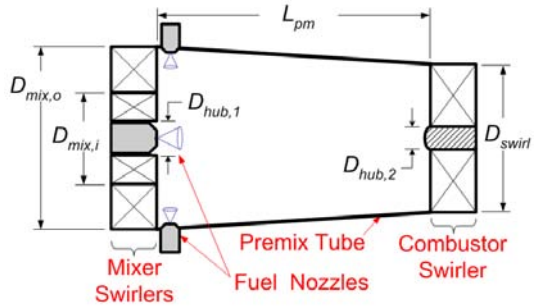

Figure 5. PREMIXER DIMENSIONS.

Injector Selection. Injector selection is a critical step in the premixer design. The injector(s) must provide a sufficiently fine mist of fuel droplets without requiring excessive fuel line pressure. Finer droplets require less time to evaporate and allow for shorter premixer tube lengths.

Nozzle selection involves choosing a nozzle type to provide the required mass flow, droplet size, and spray pattern at a specified nozzle pressure drop ΔP_n. The number of nozzles is then selected to provide favourable fuel scheduling characteristics.

Typically, nozzle manufacturers provide the mass flow or volume flow rate versus nozzle pressure relationship. For pressure nozzles:

$$Q = k\sqrt{\Delta P_n} \tag{27}$$

Another important parameter used in the premixer design is the ideal droplet discharge velocity:

$$V_n = \sqrt{\frac{2\Delta P_n}{\rho_l}} \tag{28}$$

Axial Swirler Design. Figure 6 illustrates a typical axial swirler. The axial swirlers for the premixer are designed to pass the specified mass flow $\dot{m}_{sw}$ at a desired pressure loss ΔP_{sw} while generating a certain degree of swirl. Swirler design involves selecting the type of vanes used (flat or curved), the vane angle θ_v, vane thickness t_v, vane aspect ratio AR_v, solidity σ, and inner diameter D_{hub}. From these parameters, the number of vanes n_v and the outer diameter D_{sw} are computed.

The solidity is a measure of the apparent blockage of the swirler or vane overlap. It is defined as follows:

$$\sigma = \frac{c_v \sin\theta_v}{(\pi D)/n_v} \tag{29}$$

where D is the diameter at the location the solidity is evaluated. For a solidity less than unity, the vanes do not overlap and one can see through the swirler. A solidity of unity dictates that one cannot see through the swirler yet the vanes do not overlap. The vanes completely overlap for a solidity greater than unity. Based on the experiments performed by Martin [15], a solidity between 0.8 and 1 is desirable for generating strong swirl.

The mass flow rate of air through the swirler strongly depends on the pressure loss across this device. However, relating the two is difficult due to the lack of empirical correlations. Knight and Walker [16] provided one correlation for an axial

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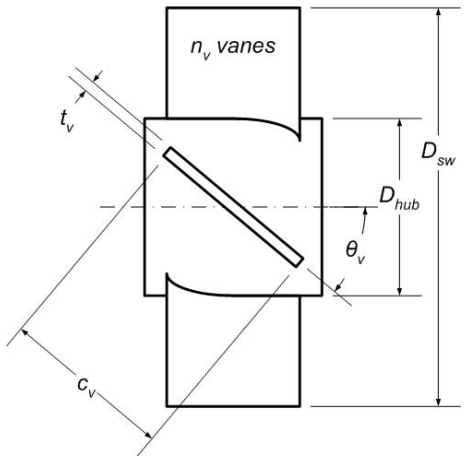

Figure 6. AXIAL SWIRLER DIMENSIONS.

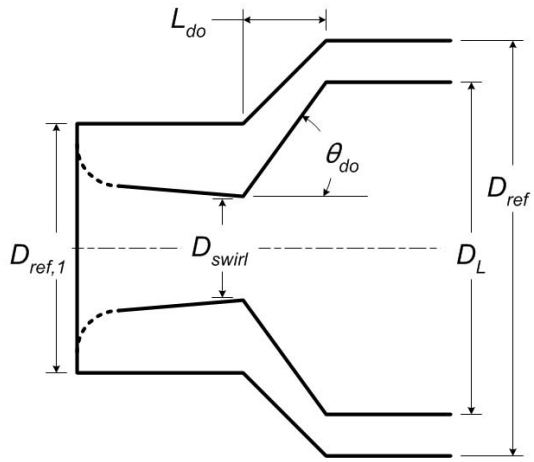

Figure 7. THE DOME REGION.

swirler admitting air into a combustion chamber:

$$\dot{m}_{sw} = \left\{ \frac{2\rho_3 \Delta P_{sw}}{K_{sw}\left[\left(\frac{\sec\theta_v}{A_{sw}}\right)^2 - \frac{1}{A_d^2}\right]} \right\}^{0.5} \tag{30}$$

where K_{sw} = 1.3 for flat vanes and 1.5 for curved vanes, and A_d is the cross-sectional area downstream of the swirler.

The swirler frontal area is defined from geometry:

$$A_{sw} = \frac{\pi}{4}\left(D_{sw}^2 - D_{hub}^2\right) + n_v\left(\frac{t_v}{\cos\theta_v}\right)\left(\frac{D_{sw} - D_{hub}}{2}\right) \tag{31}$$

The length of the vanes is determined from the aspect ratio, AR_v, and the number of vanes is fixed by the solidity:

$$c_v = \left(\frac{D_{sw} - D_{hb}}{2AR_v}\right) \tag{32}$$

$$n_v = \sigma\frac{(\pi D)}{c_v \sin\theta_v} \tag{33}$$

where AR_v is equal to the blade height divided by its length.

Swirl strength is measured by the swirl number, S_n. For axial swirlers, it is estimated by [17]:

$$S_n = \frac{2}{3}\left[\frac{1-(D_{hub}/D_{sw})^3}{1-(D_{hub}/D_{sw})^2}\right]\tan\theta_v \tag{34}$$

Lilley [17] states that strong swirl occurs when $S_n > 0.6$, which is characterized by flow recirculation.

Premixer Tube. Ideally, the premixer should be designed to ensure that complete evaporation and mixing occurs at all operating conditions without the occurrence of autoignition. However, this design considers only the higher power settings. As a result of this decision, unevaporated droplets are expected to escape the premixer at low power settings. These droplets produce small individual pilot flames inside the liner and improve the stable operating range of the combustor.

For simplicity, it is assumed that mixing occurs instantaneously and the length is based solely on considerations for evaporation. This implies that evaporation occurs at a much slower rate than mixing, which is a reasonable assumption when the temperature and pressure inside the premixer is low.

The evaporation time for the liquid fuel is estimated by modelling the spray as a single droplet with a diameter equal to the Sauter mean diameter (SMD) of the spray. A relatively simple Newtonian model was developed according to equations in [4, 18, 19] for single-component vaporization with rapid internal mixing. The evaporation model includes both the heat up and steady state evaporation periods whereby the diameter of the droplet decreases in both. Additionally, heat and mass transfer from the droplet are corrected for convection and turbulence.

The trajectory of the droplet was modelled by considering the effects of mass transfer on the drag force and the reduction in diameter as the particle evaporates. The effects of gravity, buoyancy, and turbulent dispersion were not incorporated into the model.

Autoignition and Flashback. Autoignition is most likely to occur at full power when the temperature and pressure in the premixer are highest. It can be avoided by ensuring that the mixture of fuel and air do not remain in the premixer tube for a period of time longer than the ignition delay time. A compromise between evaporation, mixedness, and autoignition must typically be made.

Wolfer [20] provided a simple means of correlating the ignition delay time with temperature and pressure:

$$t_{ig} = 0.43P^{-1.19}\exp\left[\frac{4650}{T_u}\right] \tag{35}$$

where t_{ig} is in milliseconds, P is in bars, and T_u is in degrees Kelvin.

According to Poeschl et al. [21], flashback should not be a problem if the velocity in the premixer is higher than 20 m/s.

Dome Region. The dome, illustrated by Fig. 7, is tailored to provide the desired flow pattern in the primary combustion zone. A steep wall angle angle θ_{do} of 60° was chosen based on the recommendations of Sawyer [1].

With the dome angle known, the dome length may be computed from trigonometry:

$$L_{do} = \frac{D_L - D_{swirl}}{2}\tan\theta_{do} \tag{36}$$

The diameter of the casing near the dome region, $d_{ref,1}$, is selected to minimize total pressure losses. This includes the sud-

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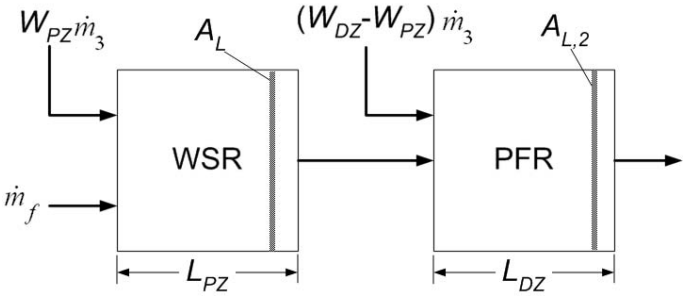

Figure 8. COMBUSTOR REACTOR MODEL.

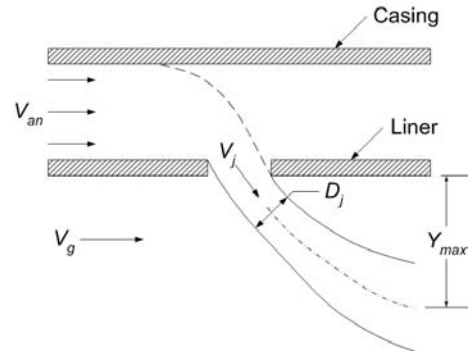

Figure 9. JET PENETRATION.

den expansion losses incurred as the flow enters the plenum and the turning losses incurred as the flow changes direction upon entering the premixer.

Combustor Length. The liner overall length is the sum of the lengths of the primary and dilution zones, illustrated in Figure 2.

The dilution zone provides the time for the hot primary zone combustion gases to mix with the cooler annulus air. The length of this zone is determined based on jet mixing considerations since the combustor must provide the downstream turbine with a specific pattern factor Q. The pattern factor is defined as:

$$Q = \frac{T_{max} - T_4}{T_4 - T_3} \tag{37}$$

where T_{max} is the maximum recorded temperature at the combustor exit. The pattern factor is estimated by modelling the dilution of the hot combustion gases with the colder jets of air admitted through the dilution holes. This is discussed later.

Unlike the dilution zone, the length of the primary zone is set to provide sufficient combustion volume to operate stably and efficiently. This value must typically be compromised to avoid an excessively large combustor with a low volumetric heat release rate.

The primary zone length is determined using reactor theory. Reactor theory allows the designer to describe the detailed evolution of the reaction within the combustor using simplified chemical reactors. The reader is directed to [7] for a mathematical description of the different types of reactors.

The combustor is modelled using a simplified network consisting of two reactors: a well-stirred reactor (WSR) and a plug-flow reactor (PFR) (Fig. 8). The WSR simulates combustion in the primary zone while the PFR simulates the dilution of the combustion products in the dilution zone. A 30-step reaction mechanism devised by Kollrack [22] for modelling the combustion of "jet fuel", represented as $C_{12}H_{23}$, was used.

Dilution Hole Design. The liner dilution holes are designed to provide the desired penetration of the admitted jets. The hole diameter D_h and number of holes n_h are determined by the available pressure drop across the liner ΔP_L, the total mass flow rate of the dilution air $\dot{m}_j$, and the mass flow rate of hot gas in the liner $\dot{m}_g$. The pressure drop across the liner is estimated from:

$$\Delta P_L = \Delta P_{3-4} - P_{diff} \tag{38}$$

The Cranfield and the NASA design methods are both commonly used to perform the design of the dilution holes. The Cranfield design method is used as it provides an optimum hole size for a given momentum flux ratio J, and selects the spacing between the adjacent holes to provide the desired $\dot{m}_j/\dot{m}_g$ ratio [9].

Hole design begins with the computation of the jet diameter D_j necessary for a given penetration distance, which is the maximum depth reached by the jets (Fig. 9). The maximum penetration distance of round air jets can be estimated by [9]:

$$Y_{max} = 1.25 D_j J^{0.5} \frac{\dot{m}_g}{\dot{m}_g + \dot{m}_j} \tag{39}$$

Lefebvre [9] suggests designing for an optimum penetration distance of $0.33D_L$ for tubular combustors.

The momentum flux ratio is defined by:

$$J = \frac{\rho_j V_j^2}{\rho_g V_g^2} \tag{40}$$

where j denotes the dilution jet while g denotes the hot gas flow.

The velocity of the issuing dilution jet and the hot combustion gases is given by:

$$V_j = \sqrt{\frac{2\Delta P_L}{\rho_3}} \tag{41}$$

$$V_g = \frac{\dot{m}_g}{\frac{\pi}{4} D_L^2 \rho_g} \tag{42}$$

$$\rho_g = \frac{P_3}{R_a T_g} \tag{43}$$

where the hot gas temperature, T_g, at the end of the primary zone is roughly equal to the flame temperature, T_{fl}.

From mass conservation, the required number of holes:

$$n_h = \frac{\dot{m}_j}{\frac{\pi}{4} D_j^2 \rho_3 V_j} \tag{44}$$

The discharge coefficient, C_d, relates the actual hole area to the effective jet flow area as the jet contracts and accelerates through the opening.

$$D_h = D_j / \sqrt{C_D} \tag{45}$$

A simple correlation for estimating the discharge coefficient

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for round sharp-edged holes is provided by Kaddah [23]:

$$C_D = \frac{1.25\,(K-1)}{[4K^2 - K(2-\beta)^2]^{0.5}} \tag{46}$$

where β is hole-to-annulus mass flow rate ratio. The coefficient K is approximated by the following correlation [23]:

$$K = 1 + 0.64\left\{2\mu^2 + \sqrt{4\mu^4 + 1.56\mu^2(4\beta - \beta^2)}\right\} \tag{47}$$

where α is the hole-to-annulus cross-sectional area ratio and $\mu = \beta/\alpha$.

Modelling the cool jet as it penetrates and mixes with the hot combustion gases is useful for verification of the hole design procedure and estimation of the combustor exit pattern factor. One empirical model for characterizing crossflowing jets is described by Holdeman et al. [24]. The model provides a detailed prediction of the jet penetration profile.

Liner Wall Temperature. Successful mechanical design requires adequate removal of heat to avoid damage of the liner. The liner contains hot gases within it that are expected to greatly exceed material limits, roughly 1100 K for nickel- and cobalt-based alloys [9].

Designing a cooling scheme to ensure adequate heat removal is facilitated by applying heat transfer theory. This allows one to predict the effects of important design parameters on the liner wall temperature. First, the wall temperature is computed excluding the effects of any supplemental cooling so that the designer may gauge the cooling requirements. A cooling scheme is designed and the liner is analyzed once again, this time including the effects of supplemental wall cooling. In this case, the cooling scheme uses ABC cooling by means of transverse turbulence promoting ribs and a TBC.

The steady state uncooled wall temperature is computed using the simple heat transfer model developed by Lefebvre and Herbert [25]. The heat transfer model uses empirical correlations to estimate the various heat fluxes and applies an energy balance to solve for the metal temperature of the liner. The heat transfer processes included are: conduction, internal and external radiation, internal and external convection. The model was updated based on the recommendations of Kretschmer and Odgers [26] for the flame luminosity.

Even though any supplementary cooling has been neglected, the results do not necessarily provide the maximum possible metal temperature. Local regions of low annulus velocity, neglected in the current model, will produce high metal temperatures above the average value computed by the model.

The heat transfer model requires an estimation for the temperature distribution of the hot gases. Since combustion is premixed, it is reasonable to assume that the temperature of the hot gases everywhere in the primary zone is equal to the flame temperature. The gas temperature is assumed to rise linearly along the dome to account for a finite rate reaction.

The gas temperature drops in the dilution zone when jets of cool annulus air are admitted into the liner. The rate of mixing between the cool jets and the hot combustion products determines how quickly the temperature drops. If the circumferential hole spacing is small enough, mixing in the near wall regions occurs quickly. Thus, the gas temperature near the liner wall in the dilution zone is approximately equal to T_4.

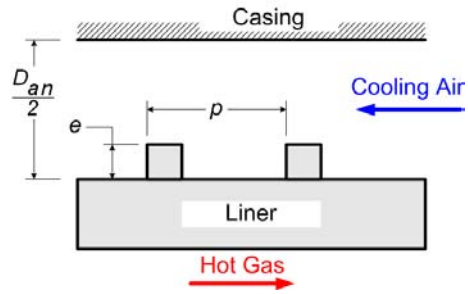

Figure 10. TRIP STRIPS.

Trip Strips. The rate of convective heat transfer from the liner wall to the annulus is augmented using transverse turbulence promoting ribs, called trip strips. These trip strips, illustrated in Fig. 10, are thin circumferential protrusions along the backside of the liner that cause the boundary layer to separate and reattach downstream. The boundary layer begins to build again after reattaching, starting from a zero thickness. Thin boundary layers and the regions of reattachment produce relatively higher heat transfer rates.

An analytical method provided by Evans and Noble [27] was used for designing the trip strips and predicting their effect on the liner wall temperature. Based on the rib height e, spacing p, and existing annulus hydraulic diameter D_{an}, the correlations provide a means for estimating the convective heat transfer coefficient on the cold side of the liner. Thus, the external convection term of the existing uncooled heat transfer model can modified to include the effects of the trip strips.

The rib height and spacing are chosen to yield the desired liner wall temperature.

Thermal Barrier Coating. Thermal barrier coatings are thin layers of refractory material that coat the inner surface of the liner. With their low thermal conductivity, they reduce the liner wall temperature by increasing the thermal resistance between the hot gas and the liner metal. Because of their low emissivity, additional reductions in wall temperature arise from the reflection of a significant portion of the incident radiation.

The reduction in wall temperature achieved from using a TBC may be calculated by modifying the heat transfer model discussed above. Another conduction term is added to the wall energy balance and the inner liner wall emissivity is updated to reflect that of the TBC material.

VERIFICATION

The design methodology was applied to a 1-MW marine gas turbine engine burning diesel fuel. To assess the success of the preliminary design procedure, the aerodynamic performance of key components at the design point was verified using computational fluid dynamics (CFD). Verification was performed by analyzing the reacting flow inside the liner and the flow inside the premixer. While a summary of the results are presented in this section, they are discussed in detail elsewhere [6].

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Combustor CFD. Analysis of the reacting flowfield inside the liner was performed using CFX-5, which is a commercially available CFD software package. The combustor flow domain was modelled using a 1/10th section consisting of 150,000 nodes and 800,000 elements. This domain was solved with the k-ε turbulence model, the combined Eddy Dissipation / Finite Rate Chemistry model, and the Differential Approximation (P1) radiation model. Finite rate chemistry was approximated by the two-step reaction mechanism for decane of Westbrook and Dryer [28]. A two-step Zeldovich mechanism was added to compute the mass fractions of thermal NO formed.

The velocity field inside the combustor, predicted using CFD, is depicted in Fig. 11. A large torroidal recirculation zone at the centre of the combustor is visible. This is typical of strongly swirling flows and indicates that the premixer blade angle is appropriate. Additionally, adequate penetration of the dilution jets can be observed. This indicates that the hole design procedures functioned properly.

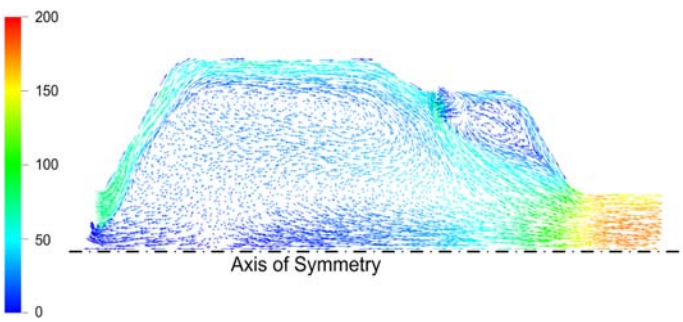

Figure 11. PREDICTED COMBUSTOR VELOCITY FIELD (m/s).

Figure 12 illustrates the temperature contours along a cross-section of the combustor, which were predicted using CFD. A comparison between these values and those computed by the preliminary design procedures is provided in Table 1. While good agreement between the predicted turbine inlet temperatures (TIT) was observed, CFD computed a slightly higher flame temperature. These discrepancies are attributed to differences in reaction mechanisms and model resolution.

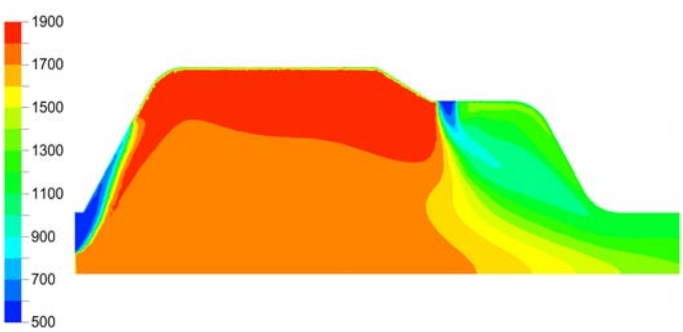

Figure 12. COMBUSTOR TEMPERATURE DISTRIBUTION (K).

Table 1. PREDICTED TEMPERATURES INSIDE COMBUSTOR (K).

	Preliminary Design	CFD
Max Flame Temp.	1803	1885
TIT	1203	1190

The emissions predicted using the preliminary design procedures and CFD are compared in Table 2. Small discrepancies between the two sets of values are observed that are primarily due to differences in reaction mechanisms (i.e. 30-step versus 2-step). These values should be regarded as crude approximations since emissions models are typically fairly inaccurate. Improving their accuracy requires the use of complex reaction schemes that are computationally intensive and difficult to obtain converged solutions.

Table 2. PREDICTED EMISSIONS (ppmvd).

	Preliminary Design	CFD
CO	3	0.5
NO	12	1
NO_2	3	N/A

Fig. 13 illustrates the liner wall temperature profiles predicted using CFD. The interface temperature refers to the temperature of the liner wall between the TBC and the liner material. The maximum temperature of the liner wall material predicted using CFD is compared to that predicted using the preliminary design procedures in Table 3. Reasonable agreement is achieved between predictions, which both indicate a satisfactory value for the peak wall temperature.

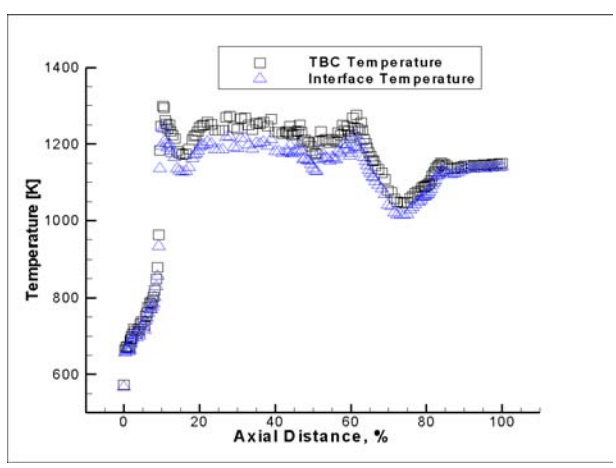

Figure 13. PREDICTED LINER WALL TEMPERATURE PROFILE.

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Table 3. PREDICTED MAXIMUM LINER WALL METAL TEMPERATURE (K).

Preliminary Design	CFD
1237	1243

Premixer CFD. Similar to the combustor, the flowfield inside the premixer was solved with CFX-5. The premixer was modelled using a 1/4 section and the domain was discretized using 125,000 nodes and 690,000 elements. No swirlers were incorporated in the simulation. The evaporating fuel spray issuing from each nozzle was modelled as a single droplet and tracked using a fully coupled evaporation model. While the effects of turbulent dispersion were neglected, the analysis included the effects of drag and heat transfer on particle behaviour. The properties of diesel fuel were approximated using the predefined "Jet A" model and the k-ε turbulence model was implemented.

Figure 14 illustrates the velocity field inside the premixer predicted using CFD. A small undesirable recirculation bubble is observed along the premixer axis. Inside this potential flameholder, however, the air/fuel ratio is too high for autoignition to occur.

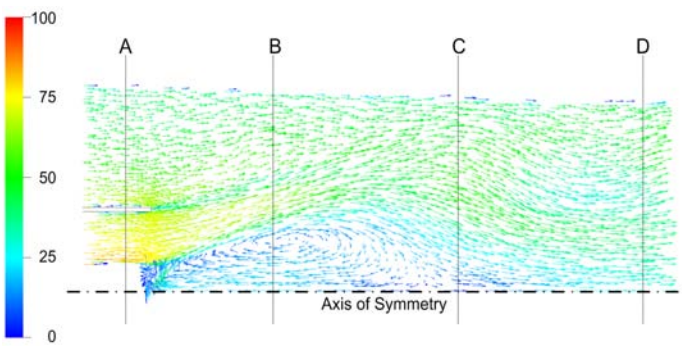

Figure 14. PREDICTED PREMIXER VELOCITY FIELD (m/s).

The contours of fuel vapour mixedness along the outlet of the premixer, which were predicted using CFD, are provided in Figure 15. Mixedness is defined as the local average mole fraction divided by the maximum inlet mole fraction. Although the droplets completely evaporate inside the premixer, multiple pockets containing higher concentrations of fuel vapour are observed. The inclusion of the effects of droplet size distributions, more realistic of actual sprays, is expected to yield more favourable results.

Table 4 provides a comparison of the evaporation distance predicted using CFD and the preliminary design procedures. A large discrepancy was observed between the predicted values. This was largely attributed to the different methods used for computing the thermodynamic and transport properties of the vapour phase surrounding the droplet. As these properties strongly affect the evaporation rates of liquid droplets, further investigation into a more accurate means for their prediction is necessary.

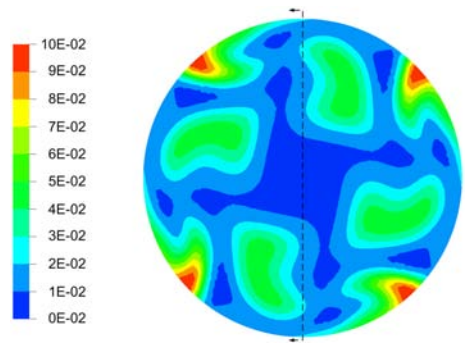

Figure 15. PREDICTED CONTOURS OF FUEL VAPOUR MIXEDNESS ALONG PREMIXER OUTLET.

Table 4. PREDICTED EVAPORATION DISTANCE FOR A 38 μm DROPLET (mm).

Preliminary Design	CFD
517	190

CLOSURE

A methodology for the preliminary design of a modern LPP gas turbine combustor was successfully developed and applied to a 1-MW marine gas turbine. The design was verified using computational fluid dynamics (CFD). Reasonable agreement between predictions from the preliminary design and CFD was achieved which indicated that the design procedures have been developed successfully.

The preliminary design procedures provided a combustor geometry that was verified to operate with high combustion efficiency and emit ultra-low concentrations of pollutants. Additionally, the procedures quickly provided a detailed definition of the geometry and an assessment of its performance. This greatly simplified the transition from preliminary to detailed design.

Verification was performed by analyzing the reacting flow inside the liner and the flow inside the premixer. While the results from CFD analysis agreed well with the predicted values for TIT, peak flame temperature, and peak wall temperature, significant discrepancies existed between emissions concentrations and droplet evaporation rates. It should be emphasized that, despite these large discrepancies, numerical analysis confirmed that the preliminary design was successful. More realistic models, in addition to experimentation, are required to improve the assessment of the preliminary design.

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